



SERVICE BULLETIN



SUBJECT: MP3000 Main Relay Board Replacements

BULLETIN NO: C 051

DATE: April 29, 2002

This bulletin recommends some checks to be performed before replacing a MP3000 Main Relay Board (MRB). There have been many MRB's replaced and returned to Thermo King with no trouble found when tested by the vendor. Under the new warranty procedures, claims for non-defective parts will not be paid and the parts returned to the submitting dealer.

There are three separate voltage outputs from the transformer:

- 29 V (wires X1 and X2) for: Compressor contactor and valve solenoids
- 28 V (wires 3 and 4) for: Supply power for the controller
- 40 V (wires 5 and 6) for: Supply power for Main Relay Board (MRB)

Symptom: MP3000 controller power up, but unit will not start

- **Check for CB2 Open Circuit (nominal 29 V)**
 - o Push bottom circuit breaker on the side of the control box next to the ON / OFF switch
 - o ATO type fuse in control box – check for continuity with ohm meter
 - o Check for output voltage of transformer, wires X1 and X2
 - o Check voltage at connector J17, pin #1 (wire X2) and pin #2 (wire 1)

Units built **before** April 2002 (032XXXXXXX)

CB2 – Control box near ON / OFF switch

Units built **after** April 2002 (042XXXXXXX)

ATO type fuse Inside control box



MGR



SALES



SERVICE

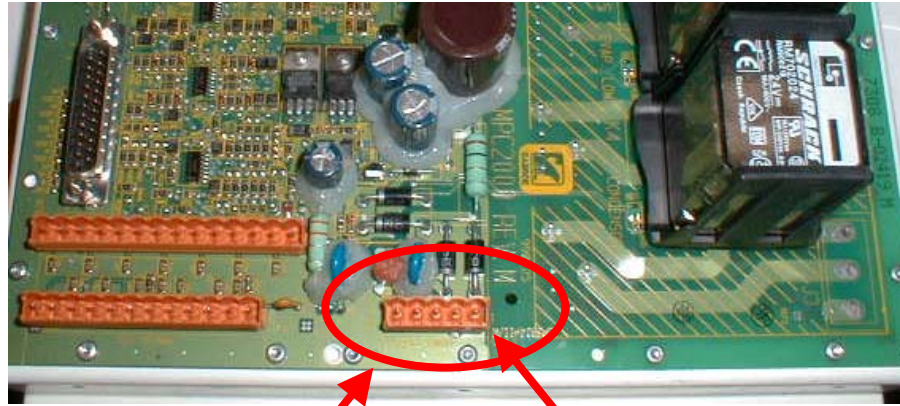


PARTS



Symptom: MRB has power, but MP3000 controller will not power up

- **Check for 28 VAC Supply to MRB**
 - o Check for output voltage of transformer, wires #3 and #4
 - o Check for voltage at ON / OFF switch. Wire #7
 - o Check voltage at connector J10, pin #2 (wire 4) and pin #3 (wire 7)



Connector J10

Pin #1

Symptom: MP3000 controller has power, but MRB will not power up

- **Check for 40 VAC supply to MRB**
 - o Check for output voltage of transformer, wires #5 and #6
 - o Check voltage at connector J10, pin #4 (wire 5) and pin #5 (wire 6)